



Travel-time reliability impacts on railway passenger demand: a revealed preference analysis

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ABSTRACT

We study the impact of travel-time reliability on trips made by railway passengers. Unlike most of the studies in this area, which make use of stated preference survey data, we use a revealed preference data set obtained by measuring the railway reliability and the number of season-ticket holders on the Dutch railway network. We make use of six travel-time reliability indicators, including the standard deviation and the 80th minus the 50th percentile of travel time. Our results indicate that the 80th minus the 50th percentile indicator best explains the fluctuations in the number of season-ticket holders. A 10% improvement of this indicator results in a 1.47% increase in the number of season-ticket holders.

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1. Introduction

The common way to analyse the benefits of transport system improvements is via travel-time savings. More recently, the unreliability of travel times has become an important element of research in this domain. A main result of research during the last decade is that travel-time reliability plays an essential role in travellers' route-choice behaviour (Bates et al., 2001; Lam and Small, 2001; Small et al., 1999; Tseng, 2008). Hence, cost-benefit analysis of transport infrastructure improvements that ignores reliability benefits may miss important aspects. Most researchers of this subject use stated preference data and focus on car transport. The present study differs from this in two respects: it is based on revealed preference methods, and it addresses the effect of reliability on railway travel demand. According to the literature in the area of customer satisfaction, travel-time reliability is one of the most important quality aspects of railway passengers (Brons and Rietveld, 2008). The more unreliable the travel time is, the higher are what are called the scheduling costs, and the less the likelihood that the train is chosen as a mode of transport. If on a certain route daily commuters are confronted with decreasing reliability, they may tend to choose a different mode of transport, if it is available.

One of the issues is how to measure reliability. As we will see, a wide range of definitions of reliability is available. In the Netherlands, the Dutch Railways (NS) measures reliability by means of arrival punctuality figures on a station level. The Dutch Ministry

of Transport defines a train as 'punctual' if it arrives at most 3 min late. Every year the railway company is held accountable for the actual punctuality in relation to the target of 87% (86.7% in 2007, NS, 2007) of trains that should be punctual (max 3 min delay). Lack of punctuality of train services and the resulting delays for the total journey play an unmistakably negative role in the evaluation of the service quality by the traveller. Also the contract between the national government and NS involves certain penalties when the reliability targets are not achieved.

Apart from the probability of a 3-min delay, there are also other ways to measure reliability such as a probability of delay longer than (say) 9 min, or the standard deviation of arrival times. As we will show, the various ways of measuring reliability are positively correlated, but correlations are sometimes surprisingly low. Therefore, we want to know which measure of reliability comes closest to the perception of travellers, and more in particular: which measure is the best predictor of actual travel behaviour. This is important, because choosing the most appropriate indicator means that the contract between the public sector and the railway operator can be based on a suitable index which provides the right incentives for the operator. To our knowledge, such a systematic comparison of various reliability indicators and their impacts on travel demand has not been carried out thus far.

This paper aims to estimate the elasticity of rail trip demand with respect to travel-time reliability for various reliability indices. The main issue addressed in this paper concerns the impact of an improvement in the reliability of travel time on railway travel, and demand in particular. Which measure of travel-time reliability can best explain variation in travel demand. A distinguishing feature of this paper is that it makes use of revealed preference data (RP), and thus reflects the actual behaviour that train

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¹ The views expressed are purely those of the author and may not in any circumstances be regarded as stating an official position of the European Commission.

travellers exhibit when they are confronted with fluctuating travel-time reliability. Bates et al. (2001) have already stated in their article about the valuation of reliability for personal travel that there is a general economic tradition to favour data that are related to observed choices. However, there are often problems with finding real choice situations with sufficient variation to allow statistically reliable estimates to be obtained at the level of detail that is required. Good studies based on revealed preference data are rare, and often only modest results are obtained.

We will address two main research questions: How large is the effect of a change in travel-time variability within the Dutch railway network on the number of season-ticket holders? And: What is the most appropriate way of defining travel-time reliability? This will be systematically explored by using a range of unreliability indicators focusing on their different performance as predictors of the travel behaviour of season-ticket holders.

This paper is structured in the following manner. In Section 2 we briefly go into the role reliability plays in transport and discuss how reliability can be defined and measured. In Section 3 data issues are examined. Section 4 provides an analysis of the descriptive statistics, and then in Section 5 we present the results of the statistical analysis of the relation between travel-time reliability and season-ticket holders. Section 6 concludes.

2. Travel-time reliability

In the literature two distinctions are made concerning the appreciation of reliability of travel times: direct appreciation on the basis of travel-time variation, and indirect appreciation on the basis of rescheduling costs (Noland and Polak, 2002). The difference here lies in the monetary valuation of the reliability. Travellers will, in general, attach a different value to early and late arrivals and departures, because they have different consequences. Most of the research literature makes a distinction between these two values by using separate terms: schedule delay early (SDE), and schedule delay late (SDL). Travel-time reliability makes the traveller incur costs in the form of uncertainty of travel time and possible trip scheduling costs. In this paper, we focus on the relationship between travel-time reliability figures and the number of season-ticket holders,² and on obtaining estimates that tell us something about the sensitivity of train travel demand to changes in travel-time reliability.

Tseng (2008) surveys the literature on the valuation of travel-time uncertainty, and finds that different definitions of (un)reliability are given. Table 1 provides an overview of a selection of relevant studies. It appears that the older studies use RP data, whereas recently SP data and combinations of RP and SP data have become more popular. Most studies address road transport demand; travel by rail is under-represented.

In the selection of studies of Tseng et al. (2008), (un)reliability is often measured as the standard deviation of the travel-time distribution (seven studies) or as the difference between the 80th (or 90th) percentile and the median of the travel-time distribution (20 studies). In this study we will make use of these two travel-time reliability indicators, in combination with several other indicators, presented in the next section.

3. Data overview

For our analysis³ we made use of data concerning the number of season-ticket holders on various origin–destination combinations on

the Dutch railway network. We included all routes that (i) do not require a transfer; and (ii) that are used by at least 11 season-ticket holders. The decision to leave out transfer passengers was based on lack of data for computing unreliability for this group. This results in 288 origin–destination routes spread over the whole Dutch railway network.

The data concerning season-ticket holders were collected for the period between January 2004 up to and including December 2007. We focus on season-ticket holders who travel on specific routes because these season-ticket holders will indeed travel on the routes for which their season ticket is valid.⁴ This is an attractive group for our purposes because they are daily commuters, and they are well informed about travel-time reliability on their route. In addition, we make use of reliability data. These reliability data were collected at the station level and provide the arrival and departure punctuality of all the train series on the Dutch railway network. These data concerning travel-time reliability were collected and made available by ProRail, the organisation in charge of the management of the rail network infrastructure. The database covers the period from June 2004 up to and including December 2007, and is aggregated by month (for 43 months). For each station and train series we have the following information with respect to travel-time reliability:

- Number of trains with {<3; 3–5; 5–7; 7–9;>9} minutes of delay on arrival (departure).
- Total number of arriving (departing) trains.
- Total number of minutes of delay on arrival (departure).
- Total number of minutes of delay on arrival (departure) for trains that arrive (depart) with more than 3 min of delay.

The data are separately available for different parts of the day (morning peak, evening peak and off-peak hours) and different parts of the week (weekdays and weekends). Here we only look at the weekdays and focus on the morning and evening peaks, since commuters mainly travel during these parts of the day and week. To compute unreliability indicators for the selected routes, we first determined which train series goes on which routes. For every route, all train series delay figures were combined into one figure for every measurement point mentioned above. The reliability data were collected for the period between June 2004 up to and including December 2004, the whole of 2005, 2006 and 2007. On the basis of the available data we will use indicators based on punctuality; indicators based on the size of the delay; and indicators based on travel-time variation. We will use six travel-time reliability indicators that have already been used in earlier research on travel-time reliability (Brons and Rietveld, 2007, 2008):

1. Percentage of delayed trains; more than 3 min delay.
2. Percentage of very delayed trains; more than 9 min of delay.
3. Average minutes of delay.
4. Average minutes of delay for delayed trains.

Standard deviation of the arrival/departure time distribution.⁵ The 80th minus the 50th percentile of the arrival/departure time distribution.⁶

The six indicators all represent the distribution of delays with one particular figure. Each indicator focuses on a particular aspect of the distribution, and, by doing so, has the limitation that it does not deal with other aspects. For example a limitation of indicator 1 is that it does not differentiate between a delay of 4 min and one of

² Season-ticket holders for a specific route.

³ For a detailed explanation of the procedures used to derive the O–D data set for the analysis see Appendix B.

⁴ This means that we exclude a group of season-ticket holders who own another type of ticket that allows them to travel on all routes in the country.

⁵ See Appendix A for calculations of the standard deviation.

⁶ See Appendix A for calculations of the 80th minus the 50th percentile.

Table 1

Overview of empirical studies on the valuation of travel-time reliability.

References	Type of study	Choice context
Small (1982)	RP	This paper studies auto commuters in the San Francisco Bay Area. For the studied commuters a regular time of arrival at work between 42.5 min early and 17.5 min late is found. A dummy variable for car-poolers is included in the choice set. The results show that it is likely that, in order to avoid congestion, many commuters, particularly single workers driving alone, are not travelling at their preferred times of day. The results provide a means to predict the extent of schedule shifts in response to a given change in congestion
Wilson (1989)	RP	This paper analyses costs to workers arising from off-peak work schedules. A discrete model of joint travel mode/work-start time choice is estimated using survey data from Singapore. The model indicates that the cost of adjusting one's work-start time six time units away from the peak starting time is comparable to the cost of one's trip being extended by six time units. Two observed modes of transport are observed: auto and motorcycle. The results of this model indicate that transportation projects, such as increases in road capacity, which may not reduce congestion levels may be beneficial if workers are able to adjust their work start-times toward the peak starting time
Lam and Small (2001)	RP	This paper studies values of time and reliability from 1998 data on the actual behaviour of auto commuters on State Route 91 in Orange County in California, where they choose between a free and a variably tolled route. For each route at each time of day and each day of the week, the distribution of travel times across different weeks is measured using loop detector data. Unreliability is represented in the best-fitting models by the difference between the 90th and the 50th percentile. The value of time was \$22.87 per hour and the value of reliability was \$15.12 per hour for men and \$31.91 for women, respectively, 72%, 48% and 101% of the sample average wage rate
Liu et al. (2004)	RP	This paper studies the value of time and reliability with reference to the actual behaviour of auto commuters on California State Route 91. The researchers found that the estimated median value of travel-time reliability is significantly higher than that of travel time. The results indicate that travellers value a reduction in travel-time variability more highly than a corresponding reduction in the actual travel time for that journey
Ghosh (2001)	SP&RP	This dissertation is about valuing the time and reliability of commuters. The researcher studied the San Diego I-15 Congestion Pricing Project, where an existing high-occupancy vehicle lane was converted to a high occupancy toll lane. Car-poolers are allowed to use the lane for free. The results show that high-income, middle-aged, home-owning, and female commuters use this tolled facility. It is notable that the value of time estimates from SP models are significantly lower than the RP estimates
Yan (2002)	SP&RP	This dissertation uses survey data sets from a road pricing experiment in the Los Angeles area to study the diversity in motorists' preference for travel time and travel-time reliability. The results show substantial heterogeneity in motorists' preferences for both travel time and travel-time reliability. And that road pricing policies when they cater for varying preferences can increase efficiency.
Bhat and Sardesai (2006)	SP&RP	This study analyses the impact of stop-making and travel-time reliability on commuters' mode choice. The researchers make use of SP and RP data, collected from a web-based commuter survey in Austin, Texas. The results show that travel-time reliability is an important variable in mode choice by commuters and needs to be considered in travel analysis. Auto, bus, rail, non-motorized mode and motorized two-wheelers are all analysed in the mode choice model
Koskenoja (1996)	SP	This dissertation is about the effect of unreliable commuting time on commuter preferences and shows how an individual's occupation can indicate how a person is going to behave concerning unreliable commuting times. The commuter trades off different elements of unreliable travel time: expected mean travel time; expected schedule delay early; and expected schedule delay late. Even when income and family composition are controlled for, occupations affect these trade-offs
Small et al. (1999)	SP	This study is about the valuation of travel-time savings and predictability in congested conditions for highway user-cost estimation. The focus is on congested travel time. In this study an SP survey was developed and conducted. The results show that a traveller values improved reliability more than twice as much as overall travel time improvements. Also individuals commuting to work places and business travellers value travel time and predictability more highly than individuals who pursue non-work-related travel
Bates et al. (2001)	SP	This study is a general review of earlier work on the progress made towards a general theory of travellers' valuation of travel-time reliability. Also an SP case study is described, where rail travellers are studied. A key finding is that punctuality is highly valued by travellers. Public transport travellers have only a limited ability to adjust their departure times, and therefore disutility is associated with unreliability per se
Hensher (2001)	SP	This study analyses the valuation of commuter travel-time saving for car drivers in six locations in New Zealand. The researchers evaluate alternative model specifications. The paper states that, with a complex disaggregation of travel time and travel cost, RP data may be inappropriate. The results show that less restrictive choice model specifications tend to produce higher mean estimates of values of time savings compared with the widely used MNL model
de Jong et al. (2003)	SP	This study presents an error components logit model for the joint choice of time of day and mode of transport. The model is estimated on SP data for car and train travellers in the Netherlands. The results indicate that time of day choice is sensitive to changes in peak travel time and cost. The researchers conclude that policies that increase these peak attributes will lead to peak spreading
Rietveld et al. (2001)	SP	This is a case study for coping with unreliability in public transport chains in the Netherlands, with a particular emphasis on delays due to missing connections in a chain with more than one element. The valuation of unreliability is estimated by means of an SP approach. The researchers found that the valuation of a certain travel time loss of 1 min is 27 eurocents, whereas the valuation of a 50% probability of a 2 min delay is 64 eurocents. A strong risk-averse attitude towards travel time can be implied
Hollander (2006)	SP	In this study, direct and indirect models for the effects of unreliability are researched. Factors affecting bus users' behaviour and attitudes towards travel-time variability are studied in the city of York, England. The results show that the influence of travel-time variability on bus users is best explained indirectly through scheduling considerations. And that a much higher penalty is placed on late arrival compared with the mean travel time than on early arrival

30 min. For indicator 2 something similar holds true: if a railway company were to reduce long delays of say 30 min so that they become 15 min, this would remain unnoticed according to this indicator. Indicator 3 focuses on an average value, but ignores the distribution of delays; for example, it does not differentiate between the situation that all trains are delayed by 1 min, or that 2 out of 10 trains are delayed by 5 min. Something similar holds for indicator 4. The standard deviation has as a potential disadvan-

tage that it treats early and late arrivals in the same way. A limitation of the 80–50% percentile is that it ignores the distribution beyond the 80% percentile. We thus conclude that the 'ideal' indicator does not exist. Therefore, the question of which reliability indicator to use in contracts cannot be solved by using *a priori* grounds. A better approach would be that the indicator is based on travellers' preferences. This is indeed the line we take in the present paper, where we infer preferences of travellers by means

Table 2

Overview of the six travel-time reliability indicators, total number of season-ticket holders, and total number of train movements for the years 2004–2007 for the 288 selected routes.

	Year			
	2004	2005	2006	2007
Total number of season-ticket holders	100	96.5	102	105.5
Total number of train movements	100	97	99	108.5
Variable				
<i>Averages of unreliability indicators:</i>				
1: Percentage of delayed trains; more than 3 min of delay (in %)	16.38	18.29	17.51	14.62
2: Percentage of very delayed trains; more than 9 min of delay (in %)	2.91	3.28	3.36	2.56
3: Average minutes of delay (in min)	1.59	1.77	1.71	1.42
4: Average minutes of delay for delayed trains (in min)	6.65	6.68	6.81	6.69
5: Standard deviation of the arrival/departure time distribution (in min)	2.63	2.75	2.78	2.53
6: 80th minus 50th percentile of the arrival/departure time distribution (in min)	1.81	1.86	1.87	1.77

of their actual response to unreliability. We then compare the six indicators as explanatory variables for actual behaviour.

This research topic of selecting the right reliability indicator is not just a theoretical issue, but has a substantial relevance for the day to day operations of railways. If criterion 1 was to be adopted by a railway company it would be stimulated to focus on reducing short delays; reducing delays of 30–15 min would not help. The railway company might therefore cancel trains that are already strongly delayed in favour of trains with short delays in order to keep the latter delays below the 3 min standard. Thus, the criterion adopted will have an impact on traffic control measures in the case of disturbances, and the same holds true for the design of timetables.

The data concerning the percentage of trains with more than 3 and 9 min of delay, and the data concerning the average minutes of delay and the average minutes of delay for delayed trains can directly be inferred from the combined punctuality database. The fifth and sixth indicators are based on the standard deviation and on the 80th minus the 50th percentile, but these figures are not directly available, and have to be calculated. The calculations are based on a number of assumptions about the travel-time distribution of the available data, and are explained more in-depth in Appendix A.

4. Analysis of the descriptive statistics

Table 2 shows the number of trains and the indexed number of season-ticket holders for the selected routes for the years 2004 up to and including 2007, with 2004 as the index year. The month and year season-ticket holders are combined in these figures. On average, 8.6% of the total number of train travellers (passenger trips) belong to the group of link-specific season-ticket holders that we are studying (KTO, 2002–2005⁷). Further about 10.1% of the train travellers have what is called an NS/OV-year season ticket that is valid for all links. These NS/OV-year season-ticket⁸ holders will probably also travel on one route frequently, but on longer routes the costs of a route-specific season ticket will be higher than the costs of a general NS/OV-year season ticket, and, are thus less attractive to purchase. Given the lack of information on the route they travel, the latter group cannot be used for our purpose. There is also a group of special season tickets, whose holders form around 2.0% of the train travellers. Altogether, the season-ticket holders group forms around 20.7% of the train travellers. An even larger group (around 24.9 per cent) is formed by the OV student ticket holders; this group

will be less affected by changing travel-time reliability since most of them do not own a car and therefore they cannot switch to an alternative mode of transport.

Table 2 shows that in 2005, compared with 2004, there was a decline in both the number of trains and in the number of season-ticket holders. In the years 2006 and 2007, on the other hand, there was an increase in both the number of trains and in the number of season-ticket holders. Concerning the travel-time reliability indicators it is noticeable that the year 2007 was one with improving figures; all of the indicators improved with respect to the previous year. In that year the Dutch Railways (NS) in collaboration with ProRail introduced a whole new timetable (officially on 10 December 2006).

Table 2 shows that at an aggregate level the number of season-ticket holders and train frequencies follow rather similar trajectories: a decrease in 2005, followed by improvements in 2006 and 2007. A rather similar pattern is also found for the overall development of unreliability: deterioration in 2005 and an improvement in the last year. In the last part of Section 5 we will also address other main drivers of the number of season-ticket holders, such as the price of tickets and the employment situation. It appears, for example, that the increase in season-ticket prices in 2005 was relatively large; employment increased rapidly in 2006 and 2007. Hence, using aggregate figures, it is impossible to tell what the relative contribution of each of these determinants is. Therefore, we adopt a different approach in the present paper by focusing on variations in the development of a large number of O–D combinations.

Table 3 shows for each of the six travel-time reliability indicators the average level and the spread of the distribution over the 288 routes⁹. The results in the first column are weighted by the number of trains that run on each of the routes. As mentioned before, the figures for departure and arrival are combined into one figure for each route. The unpunctuality (more than 3 min late), as the NS measures it, is 16.7%, according to our calculations. This figure corresponds with previous studies (e.g. 16.3% in Brons and Rietveld, 2007). The percentage of trains that are more than 9 min late is 3.03% (e.g. 3.25% in Brons and Rietveld, 2007). The average delay

⁹ A comparison with the punctuality figures for the years 2004–2007 of NS (www.prorail.com) shows that the country average of the percentage of trains delayed by more than 3 min is slightly below the average percentage on the 288 selected routes. There are three explanations for the difference between our figures and those of NS. First, our set of routes differs from that of NS. We have a selection of routes on which a minimum number of season-ticket holders travel, and we have excluded 'thinner' routes from the analysis. Those 'thinner' routes are probably to a larger degree located in rural/provincial areas, where the punctuality is likely to be higher. Secondly, we only included the punctuality figures for the morning and evening peak hours. The punctuality during peak hours is probably lower than it is during off-peak hours. And thirdly, NS focuses on arrival punctuality, while we combined both arrival and departure into one punctuality figure.

⁷ Customer Satisfaction Research of the Dutch Railways 2002–2005 (Klant tevredenheidonderzoek).

⁸ A NS/OV-year season-ticket is a season ticket valid for all trips by rail in the Netherlands (NS), during every part of the week and the day.

Table 3

Descriptive statistics concerning the travel-time reliability per route for each of the six travel-time reliability indicators (Based on 288 routes with a direct connection, for the years June–December 2004, the whole of 2005, 2006 and 2007).

	N	Mean	Range	Minimum	Maximum	68% Range
Number of trains	1152	13,072	81,136	547	81,683	
Season-ticket holders	1152	78.24	798	11	809	
1: % Delayed (>3 min)	1152	0.1670	0.44	0.01	0.45	0.104–0.233
2: % Very delayed (>9 min)	1152	0.0303	0.09	0.00	0.09	0.016–0.044
3: Average delay of all trains in minutes	1152	1.622	3.76	−0.18	3.58	1.040–2.208
4: Average delay of delayed trains in minutes	1152	6.705	6.10	4.69	10.79	5.777–7.619
5: Standard deviation in minutes	1152	2.672	4.14	0.71	4.85	2.029–3.319
6: The 80th minus the 50th percentile in minutes	1152	1.828	1.69	1.22	2.91	1.595–2.080

Table 4

Correlation between the six travel-time reliability indicators.

	% Delayed (>3 min)	% Very delayed (>9 min)	Average delay of all trains	Average delay delayed trains	Standard deviation	80th–50th percentile
% Delayed (>3 min)	1.000	0.676**	0.939**	−0.001	0.565**	0.640**
% Very delayed (>9 min)	0.676**	1.000	0.750**	0.655**	0.937**	0.624**
Average delay of all trains	0.939**	0.750**	1.000	0.117**	0.639**	0.472**
Average delay delayed trains	−0.001	0.655**	0.117**	1.000	0.790**	0.266**
Standard deviation	0.565**	0.937**	0.639**	0.790**	1.000	0.551**
The 80th minus the 50th percentile	0.640**	0.624**	0.472**	0.266**	0.551**	1.000

All correlation coefficients are based on 288 observations per year for the years June–December 2004 and the whole of 2005, 2006 and 2007.

** Correlation is significant at the 0.01 level (2-tailed).

of all the trains and all delayed trains is respectively, 1.62 and 6.71 min. Concerning the number of season-ticket holders on the selected routes, we find that there are, on average, 78 season-ticket holders, with a minimum of 11 and a maximum of 809.

5. Analyses

5.1. Correlations between reliability indicators

Table 4 shows the correlation coefficients between the six travel-time reliability indicators. We are interested in the question to what extent the indicators tell a similar story. If the correlations are all close to 1, it is rather unimportant which indicator is picked to express the reliability status on a certain link. And it would also be rather unimportant which indicator would be selected in the contract with the rail operator: the incentive to improve reliability according to a certain indicator would also improve reliability according to the other indicators as long as the mutual correlations are high. But if the correlations are low, the efforts of an operator to improve reliability according to one reliability measure may well have disappointing effects for other measures. Hence the choice of an adequate measure becomes pertinent. We find that the correlations between the indicators are lower than one might expect: they all address, in one way or another, the same concept of reliability, but the actual values obtained are rather different. In particular the indicator ‘average delay of delayed trains’ appears to be an outlier. For example, its correlation with the percentage of trains being delayed more than 3 min is very close to zero. We conclude, therefore, that the choice of a reliability measure by railway operators of public organisations is not a trivial thing.

The results of Table 4 are similar to earlier research (Brons and Rietveld, 2007) where both the indicators “Very delayed” and “Average delay of all trains” also showed the largest correlation with the unpunctuality indicator used by NS (based on the 3-min criteria).

When we compare the correlation coefficients of the relative change¹⁰ of the six travel-time reliability indicators with the relative change in the number of season-ticket holders on each route in Table 5, it shows that the sixth indicator, the 80th minus the 50th percentile, has the strongest correlation, followed by the percentage of delayed trains (>3 min), the size of the delay in minutes, and the percentage of very delayed trains (>9 min). The second indicator (standard deviation) based on travel-time variation shows a weaker correlation; and, for the indicator related to the size of the delay, the correlation is almost equal to zero. This simple correlation analysis indicates that the sixth indicator (80th minus 50th percentile) is the most promising candidate to explain variations in changes in the number of season-ticket holders.

5.2. Regressions

We use the following regression model for the analysis of changes in season-ticket holdership:

$$\frac{y_{ij(t)} - y_{ij(t-1)}}{y_{ij(t-1)}} = \alpha_t + \beta \frac{x_{ij(t)} - x_{ij(t-1)}}{x_{ij(t-1)}} + \gamma \frac{z_{ij(t)} - z_{ij(t-1)}}{z_{ij(t-1)}} + \theta \frac{y_{ij(t)}}{1000} + \omega_{2006} D_{2006} + \omega_{2007} D_{2007}, \quad (1)$$

where j and i are the origin and destination of the selected routes; x_{ij} is one of the six travel-time reliability indicators; and y_{ij} is the number of season-ticket holders. We assume a linear relationship between the relative change in the number of season-ticket holders (y_{ij}) and the relative change in travel-time reliability (x_{ij}), with the corresponding parameter β . Further, z_{ij} is the number of trains that run on the selected routes, with the corresponding frequency

¹⁰ For the relative change of the six travel-time reliability indicators and the relative change in the number of season-ticket holders, we use changes between consecutive years. Thus for 2005 the base is 2004, and for 2006 it is 2005, etc.

Table 5
Correlation coefficients of the relative change in the six travel-time reliability indicators and the dependent variable (relative change in number of season-ticket holders).

	Relative change in % delayed (>3 min)	Relative change in % very delayed (>9 min)	Relative change in average delay of all trains	Relative change in average delay delayed trains	Relative change in standard deviation	Relative change in 80th–50th percentile
Relative change in the number of season- ticket holders	–0.095**	–0.078*	–0.083*	0.005	–0.058	–0.110**

All correlation coefficients are based on 288 observations per year for the years 2005, 2006 and 2007.

* Correlation is significant at the 0.05 level (2-tailed).

** Correlation is significant at the 0.01 level (2-tailed).

Table 6
Regression results – for the relative change in the number of season-ticket holders per route under various definitions of travel-time reliability.

Explaining variables	Parameters	% Delayed (>3 min)	% Very delayed (>9 min)	Average delay of all trains	Average delay delayed trains	Standard deviation	80th–50th percentile
(Constant)	α	–0.044**	–0.044**	–0.043**	–0.045**	–0.046**	–0.040**
Relative change in reliability	β	–0.006	–0.001	–0.008	0.073	0.020	–0.147*
Relative change in the number of trains	γ	0.018*	0.018*	0.018*	0.019*	0.019*	0.019*
Number of season-ticket holders/1000	θ	0.148*	0.148*	0.147*	0.149*	0.148*	0.151*
Dummy 2006	ω_{2006}	0.092**	0.093**	0.092**	0.092**	0.094**	0.090**
Dummy 2007	ω_{2007}	0.091**	0.092**	0.090**	0.094**	0.095**	0.083**
Adjusted, R^2		0.051	0.051	0.051	0.052	0.051	0.057
N		864	864	864	864	864	864
F -test		10.306**	10.299**	10.332**	10.428**	10.336**	11.467**

All coefficients are based on 288 observations per year for the years 2005–2007.

* Coefficient is significant at the 0.05 level (2-tailed).

** Coefficient is significant at the 0.01 level (2-tailed).

parameter γ ; the parameter θ represents the influence of the number of season-ticket holders on the routes; and D_{2006} and D_{2007} are two year-dummies for the years 2006 and 2007. Note that this specification implies that we focus on the impact of route-specific variation on route-specific demand by season-ticket holders. The impact of a general improvement of reliability on route-specific demand is captured by the year dummies. The year dummies reflect changes in a large number of other factors that do not differ across all routes, such as employment conditions, season-ticket prices, and also the average development of travel-time reliability in the country as a whole. Thus, our analysis focuses on differences in the development of season-ticket holdership between routes. A much longer time series would be needed to determine the effect of overall improvements in reliability on overall ridership. The conclusion is, therefore, that the coefficients we estimate for reliability probably underestimate the total effect of reliability on demand.

To indicate if the overall regression equations are statistically significant for each of the six travel-time reliability indicators an F -test is included in the bottom row of Table 6. Overall, all regression coefficients are significant at the 0.001 level. Furthermore Table 6 shows that only the coefficient of the sixth travel-time reliability indicator (the 50th minus the 80th percentile) is significant. This indicator concerns the spread of the delay. When the spread increases, the number of season-ticket holders decreases. The results can be interpreted in the following manner: For this sixth indicator, the constant for the year 2005 is –4%, meaning that, with a constant level of the sixth indicator the demand will decrease by 4%. If the sixth indicator increases by 10%, for example, from 1.5 to 1.65, then this further decreases the demand to –5.47%. For the year 2006 there is a uniform growth of 5.0% (this means: –4% of 2005 plus 9.0%, given the 2006 dummy variable result) so that the demand will increase to 6.67% with an improvement of the sixth travel-time reliability indicator by 10%.

Earlier studies show that travellers have a higher appreciation for a decrease in the spread of the travel time delay, than for a

decrease in the average duration of a trip (Bates et al., 2001; Brons and Rietveld, 2008). The results indicate that a decrease in the spread of the travel time delay has a significant impact on demand, but also that a decrease in average delay (and thus the average travel time) has no significant impact. We also find that a positive change in the number of trains has a positive influence on the number of season-ticket holders. An increase of 10% in the number of trains results in 0.19% extra growth in the number of season-ticket holders. In European urban transportation, higher frequency elasticities (up to 0.1) are usually observed (Transtec, 2006). In addition, the number of season-ticket holders is positively related to the growth in the number of season-ticket holders. On busy routes the relative growth is stronger than on less busy routes. We conclude therefore, that the sixth indicator of travel-time reliability appears the best candidate to explain relative changes in the number of season-ticket holders.

Fig. 1 below displays the development of train ticket prices (real prices) in the Netherlands since 2004, the employment development, real GDP (after correction for inflation) and the 6th indicator (the 80th minus the 50th percentile). Despite the increase in real ticket prices for NS/OV-year season tickets, and a small increase in the price of a season ticket for a selected OD, Table 6 shows that the dummy values for 2006 and 2007 show a larger relative increase in the use of season tickets, than in 2005. A rise in employment and a rise in GDP (corrected for inflation) in 2006 and 2007 is in line with the dummy variables' higher growth values for the number of season-ticket holders.

Fig. 2 illustrates the frequency distribution of the sixth travel-time reliability indicator (the 80th minus the 50th percentile). The distribution concerns the 288 selected routes for the total time period between June 2004 and December 2007. Improving the sixth indicator implies that the distance between the 80th and the 50th percentile decreases. Consider, for example, the route Tilburg – Oss, where the sixth indicator had a value of 2.73 min in 2007. If this value improves to the mean value of the sixth

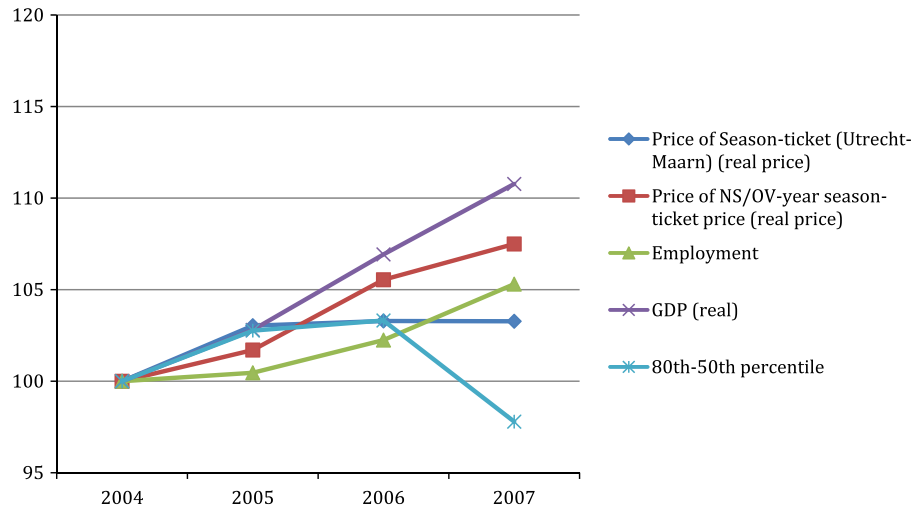


Fig. 1. The development of prices of train tickets, employment, GDP and the 6th reliability indicator since 2004.

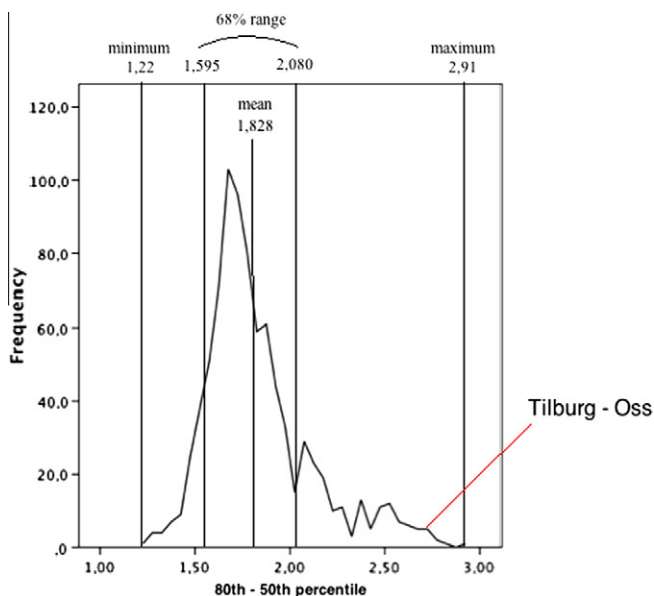


Fig. 2. Frequency distribution of the 80th–50th percentile.

indicator (1.83 min), which is an improvement of 33%, the number of season-ticket holders will increase by 4.85%.

6. Conclusions and discussion

This study has addressed the impact of travel-time reliability on the number of season-ticket holders at origin–destination routes within the Dutch rail network. The main concern of this study was the question: Does unreliability lead to an actual decrease in demand? With respect to the six travel-time reliability indicators introduced in Section 3, the results indicate that the sixth indicator (the 80th minus the 50th percentile) best reflects the fluctuations in the number of season-ticket holders. An improvement of that travel-time reliability indicator has a positive influence on the growth of the number season-ticket holders. A 10% improvement of the 80th minus the 50th percentile indicator results in a 1.47% higher growth in the number of season-ticket holders. In an earlier study (Brons and Rietveld, 2008), it was proven that both the fifth

(standard deviation) and the sixth (the 80th minus the 50th percentile) indicators reflect the reliability perception of the traveller significantly. In the present study, however, the relation with the fifth indicator was not significant. A possible explanation of the difference between the two studies is that the former was carried out for all travellers, whereas the present one is only for daily travellers; also the type of data used is different.

A further implication of this study is that, since unreliability affects travel behaviour, it apparently is a component of generalized costs, and hence should enter accessibility measures, as surveyed by among others by Banister and Berechman (2000). If unreliability were the same in all parts of the network, incorporating it in accessibility measures would not lead to new insights. However, as Fig. 2 shows, there is a rather broad variation in reliability across various routes, and this would imply that incorporating reliability might have distinct consequences for the resulting accessibility patterns.

We find that the correlations between the six reliability indicators considered are lower than one might expect. Hence the choice of a proper indicator of reliability in public transport networks is not a trivial matter. In the present study we have approached this issue of picking a proper indicator by considering its explanatory power as a determinant of changes in travel demand. This has potentially important implications for public transport operators since it would help them to formulate priorities in improving reliability, for example, by focusing on reducing extreme delays versus reducing small delays. It is also important for the public authorities that specify reliability targets for train operators. It is in the interest of all parties (public authority, concessionaire, and traveller) that the criterion specified as a target is as close as possible to traveller preferences. Our analysis shows that the reliability criterion that is currently used in the Netherlands is not the best in this list of six candidates. We recommend therefore, that the task of finding the most appropriate indicator of reliability is also studied with other data sets, in order to achieve a broader basis for drawing conclusions. One of the subjects that should receive more attention in this respect is the issue of reliability for passengers who transfer from one vehicle to another, implying that the probability of missing a connection has to be incorporated.

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Appendix A. calculation of the standard deviation and the 80th minus the 50th percentile

As discussed in Section 3, we have the following data available with respect to travel-time reliability:

- Number of trains with {<3; 3–5; 5–7; 7–9; >9} minutes of delay on arrival (departure).
- Total number of arriving (departing) trains.
- Total number of minutes of delay on arrival (departure).
- Total number of minutes of delay on arrival (departure) for trains that arrive (depart) with more than 3 min of delay.

First, we assume that the distribution of arrival and departure times are uniform within each of the five categories of delay. Standard deviation and percentiles can then be calculated based on the middle point of the intervals. With respect to the three intermediate intervals the middle points are known. However, since the first interval's lower limit and the last interval's upper limit are not known, the middle points cannot be readily calculated. The middle point of the last interval can be estimated according to the following equation:

$$X^L = \sum_{i=2}^5 M_i N_i \iff M_5 = \frac{\bar{X}^L - \sum_{i=2}^4 M_i N_i}{N_5}, \quad (2)$$

where X^L represents the total delay in minutes for delayed trains, M_i represents the middle point of interval i , N_i represents the number of trains in category i and X represents the total delay in minutes. Next, the middle point of the first interval can be calculated as follows:

$$X = \sum_{i=1}^5 M_i N_i \iff M_1 = \frac{\bar{X} - \sum_{i=2}^5 M_i N_i}{N_1}. \quad (3)$$

Based on the estimated and calculated middle points, the standard deviation can be computed as follows:

$$SD = \sqrt{\frac{\sum_{i=1}^5 (M_i - \bar{X})^2 N_i}{(\sum_{i=1}^5 N_i) - 1}}. \quad (4)$$

The 50th and the 80th percentile can be both calculated as follows:

$$P^{perc} = \left(M_1 - \frac{1}{2}R_1\right) + \sum_{i=1}^{k-1} R_i + \frac{(perc - \sum_{i=1}^{k-1} S_i)}{S_k} R_k, \quad (5)$$

where $perc$ denotes the percentile to be calculated; R_i represents the bandwidth of interval i ; k represents the interval within which the percentile is located; and S_i represents the percentage of trains in interval i .

Appendix B. procedures used to derive the origin–destination data set

The data is separately available for different parts of the day (morning peak (07:00–08:59), evening peak (16:00–18:59) and off-peak hours) and different parts of the week (weekends and weekdays). We only look at the data of the weekdays and focus on the morning and evening peaks. This because we assume that season-ticket holders are commuters who travel during these parts of the week and day. The database covers the period from June 2004 up to and including December 2007, and is aggregated by

month (for 43 months). For each station and train series we have the following information with respect to travel-time reliability:

- Number of trains with {<3; 3–5; 5–7; 7–9; >9} minutes of delay on arrival (departure).
- Total number of arriving (departing) trains.
- Total number of minutes of delay on arrival (departure).
- Total number of minutes of delay on arrival (departure) for trains that arrive (depart); with more than 3 min of delay.

Other data that are available are the number of season-ticket holders per origin and destination combination on all routes within the Netherlands, and data on train series on the Dutch railways. To reach the point where there is one punctuality figure for each route, we first determine which train series¹¹ go on which routes. This is done because the punctuality data is recorded by train series at station level. Thus every passing train is entered into the data set by arrival and departure time (punctuality) and by train series number. When a train series arrives and departs at both origin and destination in a specific O–D combination, then the arrival and departure punctuality at the origin and destination are used in the final data set for analysis.

For every route all the train series travel time (un)reliability figures are combined into one figure. Here no distinction is made between direction on the specific O–D or departure and arrival travel time (unreliability). This results into one travel time (un)reliability figure per year per O–D, for every of the six travel-time reliability indicators used in the analysis of this study.

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¹¹ The number of the train series specifies the O–D combination, the direction and the time of day.

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